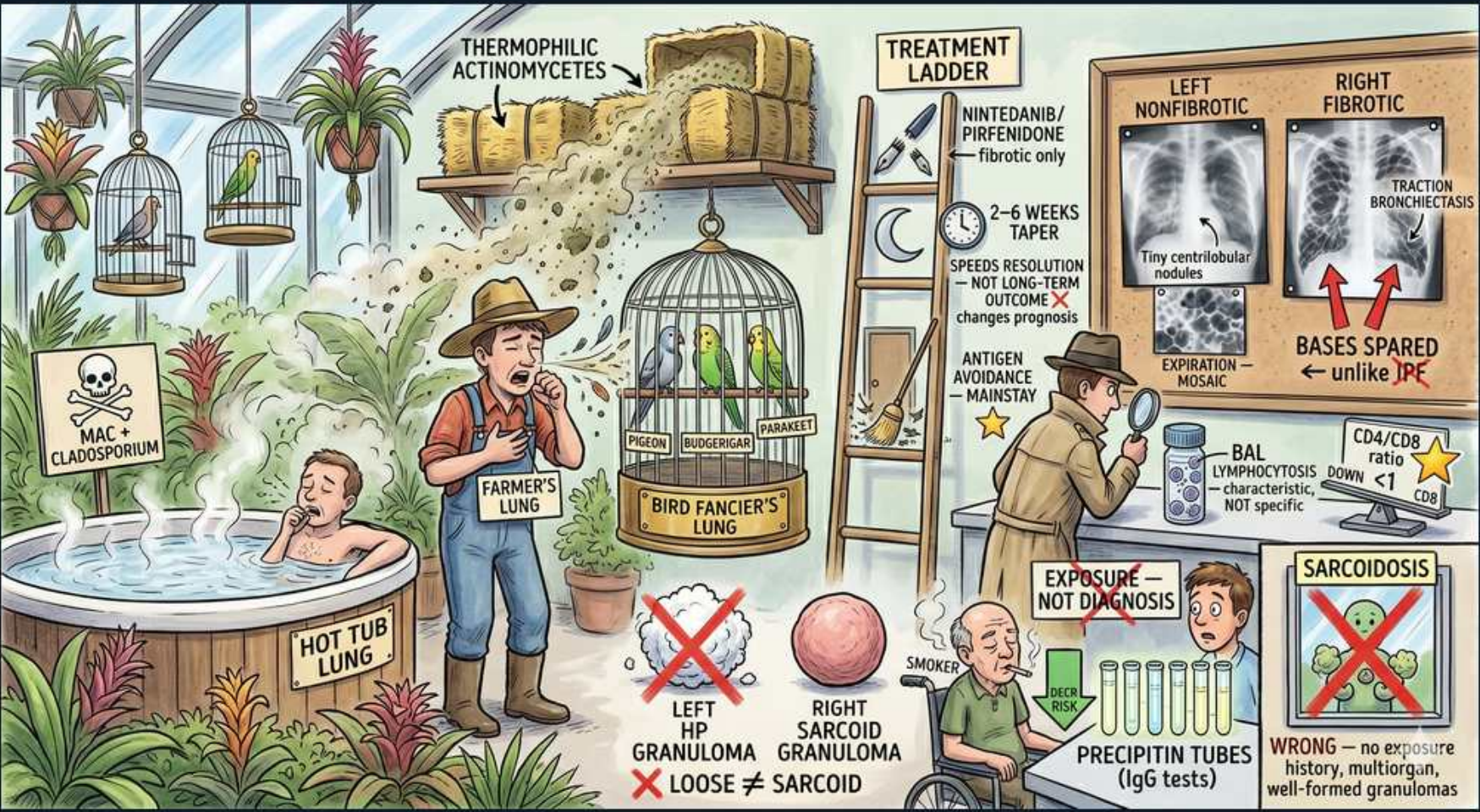


Hypersensitivity Pneumonitis — Exposures, Imaging, Dx & Treatment



1. Thermophilic actinomycetes

WHAT YOU SEE

Moldy hay pile labeled THERMOPHILIC ACTINOMYCETES

WHAT IT ENCODES

Thermophilic actinomycetes growing in moldy hay, silage, and compost are the classic antigen of farmer's lung, the prototypical hypersensitivity pneumonitis (HP). HP is an immune-mediated reaction to inhaled organic antigens, combining a type III (immune-complex, with serum IgG precipitins) acute response and a type IV (T-cell/granulomatous) delayed response. The exposure history is the single most important diagnostic clue.

BOARD TRAP

WRONG: 'HP from moldy hay is an IgE-mediated (type I) allergic/asthmatic reaction.' HP is NOT type I/IgE-mediated — it is type III (immune complexes, IgG precipitins) plus type IV (T-cell granulomatous). The discriminator: no eosinophilia/IgE, but positive serum precipitins and BAL lymphocytosis; eosinophilic/IgE findings point to asthma or ABPA instead.

2. Farmer's lung

WHAT YOU SEE

Man in overalls coughing tagged FARMER'S LUNG

WHAT IT ENCODES

Farmer's lung is the prototypical HP — ask about hay, silage, grain, and barn/livestock work. Acute episodes occur 4–8 hours after heavy antigen exposure with fever, chills, cough, and dyspnea that mimic an infection but recur with re-exposure and improve when away from the farm. Chronic low-level exposure can instead cause insidious fibrosis.

BOARD TRAP

WRONG: 'A farmer with recurrent fever, cough, and infiltrates 4–8h after barn work has recurrent pneumonia — treat with antibiotics.' The temporal link to antigen exposure (symptoms hours after work, resolving away from the farm) points to acute HP, not infection. Repeated 'pneumonia' courses that track with exposure and clear on weekends/vacation is the discriminator — give antigen avoidance, not more antibiotics.

3. Bird fancier's lung

WHAT YOU SEE

Birdcages with parrots, tag BIRD FANCIER'S LUNG (pigeon/budgerigar/parakeet)

WHAT IT ENCODES

Bird fancier's (bird breeder's) lung is HP from avian proteins in droppings, feathers, and serum — pigeons, budgerigars, parakeets, parrots. Also ask about down/feather bedding and pillows. It is one of the most common and important causes of chronic HP and a frequent cause of unexplained interstitial lung disease in pet-bird owners.

BOARD TRAP

WRONG: 'No pet birds in the home rules out bird fancier's lung.' Exposure can be occult — feather/down duvets and pillows, a neighbor's pigeons, or even prior bird ownership with persistent antigen in the home can drive disease. Failing to probe for hidden avian antigen sources (down bedding, past birds, environmental contamination) misses the diagnosis.

4. Hot tub lung (MAC)

WHAT YOU SEE

Steaming hot tub labeled HOT TUB LUNG

WHAT IT ENCODES

Hot tub lung is an HP-like granulomatous response to aerosolized Mycobacterium avium complex (MAC) from warm, recirculated water (indoor hot tubs/spas). Patients present with cough, dyspnea, and fever; HRCT shows centrilobular nodules and mosaic attenuation. Management centers on removing the exposure (drain/avoid the tub); steroids ± antimycobacterials are used in selected cases.

BOARD TRAP

WRONG: 'Hot tub lung is a true MAC pulmonary infection requiring full anti-MAC antibiotic therapy.' It is primarily a hypersensitivity/granulomatous reaction to the organism rather than invasive infection — the mainstay is antigen avoidance (stop using the tub), with antimycobacterials reserved for select/severe cases. Treating it as standard disseminated/pulmonary MAC infection is the trap.

5. MAC + Cladosporium

WHAT YOU SEE

Skull-and-crossbones placard MAC + CLADOSPORIUM

WHAT IT ENCODES

Water systems and damp environments aerosolize both Mycobacterium avium complex and molds such as Cladosporium, which can trigger HP. This links the 'hot tub lung' (MAC) and other water/mold-related HP (humidifier lung, sauna taker's lung) under the theme of contaminated water aerosols. Identifying and removing the contaminated water source is therapeutic.

BOARD TRAP

WRONG: 'HP is always caused by a single classic antigen like hay or birds.' Many HP cases come from indoor water/mold sources (humidifiers, AC units, hot tubs — MAC, Cladosporium, thermophilic actinomycetes). Anchoring only on farm/bird antigens and not investigating the home/water environment misses these increasingly common causes.

6. Antigen avoidance #1

WHAT YOU SEE

Star-marked banner ANTIGEN AVOIDANCE MAINSTAY

WHAT IT ENCODES

Antigen avoidance is the #1 treatment for HP and the only intervention that changes long-term prognosis — identify and permanently remove the exposure (rehome birds, remove down bedding, remediate mold, change work practices/use respirators). Early complete avoidance in nonfibrotic disease can be curative; persistent exposure drives progression to fibrosis.

BOARD TRAP

WRONG: 'Steroids are the definitive treatment for HP.' Corticosteroids only speed symptomatic recovery and do not alter long-term outcome — antigen AVOIDANCE is what changes prognosis. Relying on steroids while the patient keeps the bird or stays in the contaminated environment leads to ongoing injury and fibrosis. Removal of antigen comes first.

7. Centrilobular nodules

WHAT YOU SEE

Left CXR LEFT NONFIBROTIC with 'tiny centrilobular nodules'

WHAT IT ENCODES

Nonfibrotic (acute/subacute) HP on HRCT shows poorly-defined centrilobular ground-glass nodules in an upper/mid-zone predominance, often with mosaic attenuation. These reflect small-airway-centered inflammation. This pattern, with the right exposure history and BAL lymphocytosis, supports the diagnosis without requiring biopsy.

BOARD TRAP

WRONG: 'Centrilobular ground-glass nodules are nonspecific and don't help distinguish HP.' In the correct clinical context (antigen exposure, BAL lymphocytosis), upper/mid-zone centrilobular nodules with mosaic attenuation and air-trapping are quite suggestive of nonfibrotic HP. Dismissing this combination ignores a high-yield imaging clue that helps avoid biopsy.

8. Mosaic / air-trapping

WHAT YOU SEE

Label EXPIRATION MOSAIC under left film

WHAT IT ENCODES

Mosaic attenuation with expiratory air-trapping is a hallmark HRCT finding in HP, reflecting small-airway (bronchiolar) obstruction. The classic 'three-density/headcheese sign' — coexisting ground-glass, normal lung, and air-trapped (lucent) lobules — is highly suggestive of HP. Expiratory images are needed to demonstrate the trapping.

BOARD TRAP

WRONG: 'Mosaic attenuation on inspiratory images alone confirms air-trapping.' True air-trapping must be shown on EXPIRATORY HRCT (the lucent regions fail to increase in attenuation on expiration). Mosaic perfusion from vascular/inspiratory causes can mimic it; omitting expiratory imaging is the trap that leads to misinterpretation.

9. Fibrotic + bases spared

WHAT YOU SEE

Right film RIGHT FIBROTIC, traction bronchiectasis, BASES SPARED unlike IPF

WHAT IT ENCODES

Chronic fibrotic HP shows reticulation, traction bronchiectasis, and sometimes honeycombing, but with relative SPARING of the lung BASES and upper/mid-zone or diffuse predominance — the opposite of IPF, which is basal- and subpleural-predominant. Coexisting mosaic attenuation/air-trapping within fibrosis favors HP over IPF.

BOARD TRAP

WRONG: 'Basal-predominant fibrosis with honeycombing equals IPF.' IPF is basal/subpleural-predominant, but chronic HP fibroses with relative basal SPARING and upper/mid-zone predominance plus air-trapping. The discriminator is zonal distribution and air-trapping — calling upper-zone, base-sparing fibrosis 'IPF' (and missing the antigen history) is the classic trap.

10. BAL lymphocytosis

WHAT YOU SEE

BAL tubes tag BAL LYMPHOCYTOSIS — characteristic NOT specific

WHAT IT ENCODES

Bronchoalveolar lavage in HP shows marked lymphocytosis (often >30–50%) — a characteristic supportive finding, but NOT specific (also seen in sarcoidosis, drug reactions, NSIP). A normal BAL lymphocyte count makes active HP less likely. BAL helps support HP and exclude infection/malignancy in the right clinical setting.

BOARD TRAP

WRONG: 'BAL lymphocytosis is diagnostic of HP.' It is characteristic but not specific — sarcoidosis, NSIP, and drug-induced pneumonitis also show lymphocytosis. Diagnosis of HP requires integrating exposure history, imaging, BAL, and sometimes biopsy; treating lymphocytosis as proof of HP overcalls the diagnosis.

11. Low CD4/CD8 ratio

WHAT YOU SEE

CD4/CD8 RATIO with DOWN arrow

WHAT IT ENCODES

HP classically lowers the BAL CD4/CD8 ratio (CD8 lymphocyte predominance), reflecting its type IV T-cell-mediated component. This contrasts with sarcoidosis, which classically RAISES the CD4/CD8 ratio. The ratio is a helpful (though variable) adjunct when distinguishing granulomatous interstitial diseases.

BOARD TRAP

WRONG: 'A high BAL CD4/CD8 ratio supports HP.' That is backwards — HP classically gives a LOW (CD8-predominant) ratio, whereas a HIGH CD4/CD8 ratio favors SARCOIDOSIS. Mixing up the direction of the ratio is a high-yield trap that flips the differential between HP and sarcoid.

12. Precipitin tubes (IgG)

WHAT YOU SEE

Test tubes labeled PRECIPITIN TUBES (IgG tests), EXPOSURE NOT DIAGNOSIS

WHAT IT ENCODES

Serum precipitins are antigen-specific IgG antibodies marking the type III mechanism; their presence documents EXPOSURE and sensitization, not active disease. They support the diagnosis when positive against a suspected antigen but lack sensitivity and specificity on their own. Their main value is identifying/confirming the offending antigen.

BOARD TRAP

WRONG: 'Positive serum precipitins confirm hypersensitivity pneumonitis.' Precipitins indicate exposure/sensitization, NOT disease — many exposed but healthy people (e.g., farmers, bird keepers) have positive IgG without HP, and patients with HP can be precipitin-negative. Using a positive precipitin as proof of disease overdiagnoses HP.

13. Non-caseating granulomas

WHAT YOU SEE

Pink loose granuloma LEFT HP GRANULOMA — LOOSE ≠ SARCOID

WHAT IT ENCODES

Histology of HP shows poorly-formed, loose, non-caseating granulomas with cellular bronchiolocentric (airway-centered) lymphocytic inflammation and occasional giant cells — reflecting the combined type III and type IV reaction. The bronchiolocentric distribution and loose granulomas help separate HP from sarcoidosis.

BOARD TRAP

WRONG: 'Non-caseating granulomas on biopsy mean sarcoidosis.' Both HP and sarcoid have non-caseating granulomas, but HP's are LOOSE/poorly-formed and bronchiolocentric whereas sarcoid's are TIGHT/well-formed along lymphatics. The granuloma morphology and distribution (plus exposure history) discriminate them — assuming non-caseating = sarcoid is the trap.

14. Sarcoid granuloma

WHAT YOU SEE

Tight pink ball RIGHT SARCOID GRANULOMA

WHAT IT ENCODES

Sarcoid granulomas are well-formed, tight, compact non-caseating granulomas distributed along lymphatic routes (peribronchovascular, subpleural, septal). This morphology contrasts with the loose, airway-centered granulomas of HP and is a key histologic discriminator between the two granulomatous interstitial diseases.

BOARD TRAP

WRONG: 'Tight well-formed granulomas could equally be HP.' Tight, compact, lymphangitic granulomas favor SARCOIDOSIS; HP shows loose, poorly-formed, bronchiocentric granulomas. Combined with a HIGH CD4/CD8 ratio and absent exposure history, tight granulomas point to sarcoid — equating them with HP is the trap.

15. Sarcoidosis = WRONG

WHAT YOU SEE

Boxed SARCOIDOSIS crossed out: no exposure history, multiorgan, well-formed granulomas

WHAT IT ENCODES

Key differentiator: HP requires an identifiable inhaled antigen EXPOSURE and is lung-centered, whereas sarcoidosis is a multiorgan disease (hilar lymphadenopathy, skin, eyes, heart, hypercalcemia) with NO causative exposure, well-formed granulomas, and a high BAL CD4/CD8 ratio. Exposure history is the pivot that favors HP.

BOARD TRAP

WRONG: 'Bilateral hilar adenopathy with granulomas and no clear exposure is HP.' That picture — multiorgan involvement, bilateral hilar adenopathy, high CD4/CD8, tight granulomas, no antigen — is SARCOIDOSIS. HP needs an exposure history and is airway-centered with low CD4/CD8. Choosing HP without an exposure and with multiorgan/hilar findings is the trap.

16. Antifibrotics (fibrotic only)

WHAT YOU SEE

Banner NINTEDANIB/PIRFENIDONE — fibrotic only

WHAT IT ENCODES

Antifibrotics (nintedanib, an intracellular tyrosine kinase inhibitor; pirfenidone) are reserved for fibrotic/progressive chronic HP that continues to decline despite antigen avoidance and immunosuppression. They slow the rate of FVC decline in progressive pulmonary fibrosis but do not reverse established fibrosis. Antigen avoidance remains the foundation even when antifibrotics are added.

BOARD TRAP

WRONG: 'Start nintedanib/pirfenidone for acute or nonfibrotic HP.' Antifibrotics have NO role in acute/nonfibrotic HP — those respond to antigen removal ± steroids. Antifibrotics are only for progressive FIBROTIC disease. Using them in inflammatory/nonfibrotic HP (or instead of antigen avoidance) is the trap.

17. Steroid taper 2-6 wk

WHAT YOU SEE

Ladder + moon icon 2-6 WEEKS TAPER, speeds resolution not long-term outcome

WHAT IT ENCODES

Corticosteroids (e.g., prednisone ~0.5 mg/kg/day, tapered over weeks to a few months) speed symptomatic and physiologic recovery in acute/subacute or symptomatic HP but do NOT change the long-term outcome — that is determined by antigen avoidance. Reserve steroids for significantly symptomatic, hypoxemic, or progressive disease, watching for steroid adverse effects.

BOARD TRAP

WRONG: 'A good steroid response means the patient can continue antigen exposure.' Steroids only hasten recovery; they do not protect against ongoing injury from continued exposure, and improvement on steroids while still exposed gives false reassurance. The prognosis-changing step is permanent antigen avoidance, not the steroid course.